6.7 Solution: (a) Let us consider the average current density of the n-th molecule as in Eq. (6.94). Following the same procedure as for the charge density, we can Taylor expand the expression as

$$
\begin{aligned}
\langle \mathbf{j}_n(\mathbf{x},t)\rangle &= \sum_{j(n)} q_j(\mathbf{v}_n+\mathbf{v}_{jn})\left[f(\mathbf{x}-\mathbf{x}_n)-\mathbf{x}_{jn}\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)+\frac{1}{2}\sum_{\alpha\beta}(\mathbf{x}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\frac{\partial^2 f(\mathbf{x}-\mathbf{x}_n)}{\partial x_\alpha\partial x_\beta}\right]\\
&= \mathbf{v}_n\left[q_n f(\mathbf{x}-\mathbf{x}_n)-\mathbf{p}_n\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)+\frac{1}{6}\sum_{\alpha\beta}(Q'_n)_{\alpha\beta}\frac{\partial^2 f(\mathbf{x}-\mathbf{x}_n)}{\partial x_\alpha\partial x_\beta}\right] \qquad (1)\\
&+ \sum_{j(n)} q_j\left[\mathbf{v}_{jn}f(\mathbf{x}-\mathbf{x}_n)-\mathbf{v}_{jn}[\mathbf{x}_{jn}\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)]+\frac{1}{2}\mathbf{v}_{jn}\sum_{\alpha\beta}(\mathbf{x}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\frac{\partial^2 f(\mathbf{x}-\mathbf{x}_n)}{\partial x_\alpha\partial x_\beta}\right].
\end{aligned}
$$

Here, we have used the following definitions,

$$
q_n=\sum_{j(n)}q_j,\quad \mathbf{p}_n=\sum_{j(n)}q_j\mathbf{v}_n,\quad (Q'_n)_{\alpha\beta}=3\sum_{j(n)}q_j(\mathbf{x}_{jn})_\alpha(\mathbf{x}_{jn})_\beta,
$$

for molecular charge, dipole moment, and quadrupole moment, respectively.

It is straightforward to see that

$$
q_n\mathbf{v}_n f(\mathbf{x}-\mathbf{x}_n)=\langle q_n\mathbf{v}_n\delta(\mathbf{x}-\mathbf{x}_n)\rangle.
$$

For the dipole moment term,

$$
\begin{aligned}
-\mathbf{v}_n\Big(\mathbf{p}_n\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)\Big) &= \mathbf{p}_n\Big(\mathbf{v}_n\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)\Big)-\mathbf{v}_n\Big(\mathbf{p}_n\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)\Big)-\mathbf{p}_n\Big(\mathbf{v}_n\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)\Big)\\
&= \sum_\alpha\frac{\partial}{\partial x_\alpha}\left[\Big(\mathbf{p}_n(\mathbf{v}_n)_\alpha-\mathbf{v}_n(\mathbf{p}_n)_\alpha\Big)f(\mathbf{x}-\mathbf{x}_n)\right]+\mathbf{p}_n\frac{\partial f(\mathbf{x}-\mathbf{x}_n)}{\partial t}\\
&= \sum_\alpha\frac{\partial}{\partial x_\alpha}\left[\Big(\mathbf{p}_n(\mathbf{v}_n)_\alpha-\mathbf{v}_n(\mathbf{p}_n)_\alpha\Big)f(\mathbf{x}-\mathbf{x}_n)\right]+\frac{\partial}{\partial t}\Big[\mathbf{p}_n f(\mathbf{x}-\mathbf{x}_n)\Big]\\
&\qquad -\sum_{j(n)}q_j\mathbf{v}_{jn}f(\mathbf{x}-\mathbf{x}_n)\\
&= \sum_\alpha\frac{\partial}{\partial x_\alpha}\Big\langle\Big(\mathbf{p}_n(\mathbf{v}_n)_\alpha-\mathbf{v}_n(\mathbf{p}_n)_\alpha\Big)\delta(\mathbf{x}-\mathbf{x}_n)\Big\rangle+\frac{\partial}{\partial t}\Big\langle\mathbf{p}_n\delta(\mathbf{x}-\mathbf{x}_n)\Big\rangle\\
&\qquad -\sum_{j(n)}q_j\mathbf{v}_{jn}f(\mathbf{x}-\mathbf{x}_n),
\end{aligned}
$$

where we can see that the last term exactly cancels the first term in the second line of Eq. (1). We can perform the similar calculation for the quadrupole moment, the third term in the first line of Eq. (1),

$$
\begin{aligned}
&\frac{1}{6}\mathbf{v}_n\sum_{\alpha\beta}(Q'_n)_{\alpha\beta}\frac{\partial^2 f(\mathbf{x}-\mathbf{x}_n)}{\partial x_\alpha\partial x_\beta}=\frac{1}{2}\mathbf{v}_n\sum_{j(n)}\sum_{\alpha\beta}q_j(\mathbf{x}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\frac{\partial^2 f(\mathbf{x}-\mathbf{x}_n)}{\partial x_\alpha\partial x_\beta}\\
=\ &\frac{1}{2}\sum_{\alpha\beta}\frac{\partial^2}{\partial x_\alpha\partial x_\beta}\sum_{j(n)}q_j\left(\Big[(\mathbf{x}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\mathbf{v}_n-(\mathbf{x}_{jn})_\alpha\mathbf{x}_{jn}(\mathbf{v}_n)_\beta+(\mathbf{x}_{jn})_\alpha\mathbf{x}_{jn}(\mathbf{v}_n)_\beta\Big]f(\mathbf{x}-\mathbf{x}_n)\right)
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{6}\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial^2}{\partial x_\beta\partial x_\gamma}\left[\Big((\mathbf{v}_n)_\alpha(Q'_n)_{\beta\gamma}-(\mathbf{v}_n)_\gamma(Q'_n)_{\alpha\beta}\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
&\qquad -\frac{1}{6}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[(Q'_n)_{\alpha\beta}\frac{\partial f(\mathbf{x}-\mathbf{x}_n)}{\partial t}\right]\\
&= \frac{1}{6}\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial^2}{\partial x_\beta\partial x_\gamma}\left[\Big((\mathbf{v}_n)_\alpha(Q'_n)_{\beta\gamma}-(\mathbf{v}_n)_\gamma(Q'_n)_{\alpha\beta}\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
&\qquad -\frac{\partial}{\partial t}\left[\frac{1}{6}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\Big((Q'_n)_{\alpha\beta}f(\mathbf{x}-\mathbf{x}_n)\Big)\right]\\
&\qquad +\frac{1}{2}\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta+(\mathbf{x}_{jn})_\alpha(\mathbf{v}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
&= -\frac{1}{6}\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial^2}{\partial x_\beta\partial x_\gamma}\left\langle\Big((Q'_n)_{\alpha\beta}(\mathbf{v}_n)_\gamma-(Q'_n)_{\gamma\beta}(\mathbf{v}_n)_\alpha\Big)\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle\\
&\qquad -\frac{\partial}{\partial t}\left\langle\frac{1}{6}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\Big((Q'_n)_{\alpha\beta}f(\mathbf{x}-\mathbf{x}_n)\Big)\right\rangle\\
&\qquad +\frac{1}{2}\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta+(\mathbf{x}_{jn})_\alpha(\mathbf{v}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right],
\end{aligned}
$$

where the last term will be combined with the second term of the second line in Eq. (1) to form the magnetic moment,

$$
\begin{aligned}
&\frac{1}{2}\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta+(\mathbf{x}_{jn})_\alpha(\mathbf{v}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right]-\sum_{j(n)}q_j\mathbf{v}_{jn}[\mathbf{x}_{jn}\cdot\nabla f(\mathbf{x}-\mathbf{x}_n)]\\
=\;&\frac{1}{2}\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta+(\mathbf{x}_{jn})_\alpha(\mathbf{v}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
&\quad -\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
=\;&\frac{1}{2}\sum_{j(n)}q_j\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\Big((\mathbf{x}_{jn})_\alpha(\mathbf{v}_{jn})_\beta-(\mathbf{v}_{jn})_\alpha(\mathbf{x}_{jn})_\beta\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
=\;&\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\epsilon_{\alpha\beta\gamma}\Big(\sum_{j(n)}\frac{q_j}{2}\left(\mathbf{x}_{jn}\times\mathbf{v}_{jn}\right)_\gamma\Big)f(\mathbf{x}-\mathbf{x}_n)\right]\\
=\;&\nabla\times\Big[\mathbf{m}_nf(\mathbf{x}-\mathbf{x}_n)\Big]=\nabla\times\Big\langle\mathbf{m}_n\delta(\mathbf{x}-\mathbf{x}_n)\Big\rangle,
\end{aligned}
$$

where we define the molecular magnatic moment as

$$
\mathbf{m}_n=\sum_{j(n)}\frac{q_j}{2}\left(\mathbf{x}_{jn}\times\mathbf{v}_{jn}\right).
$$

Putting everything together, we will have

$$\begin{aligned}
\langle \mathbf{j}_n(\mathbf{x},t)\rangle &= \langle q_n\mathbf{v}_n\delta(\mathbf{x}-\mathbf{x}_n)\rangle + \frac{\partial}{\partial t}\left\langle \mathbf{p}_n\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle - \frac{\partial}{\partial t}\left\langle \frac{1}{6}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\Big((Q'_n)_{\alpha\beta}\delta(\mathbf{x}-\mathbf{x}_n)\Big)\right\rangle \\
&+ \nabla\times\left\langle \mathbf{m}_n\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle \\
&+ \sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left\langle \Big((\mathbf{p}_n)_\alpha(\mathbf{v}_n)_\beta-(\mathbf{v}_n)_\alpha(\mathbf{p}_n)_\beta\Big)\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle \\
&- \frac{1}{6}\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial^2}{\partial x_\beta\partial x_\gamma}\left\langle \Big((Q'_n)_{\alpha\beta}(\mathbf{v}_n)_\gamma-(Q'_n)_{\gamma\beta}(\mathbf{v}_n)_\alpha\Big)\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle,
\end{aligned}$$

and the contribution from all the molecules is

$$\mathbf{j}_{\text{bound}}(\mathbf{x},t) = \sum_{n(\text{molecules})}\langle \mathbf{j}_n(\mathbf{x},t)\rangle.$$

The summation over all molecules for the dipole and the quadrupole terms combined gives

$$\begin{aligned}
&\sum_{n(\text{molecules})}\left[\frac{\partial}{\partial t}\left\langle \mathbf{p}_n\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle - \frac{\partial}{\partial t}\left\langle \frac{1}{6}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\Big((Q'_n)_{\alpha\beta}\delta(\mathbf{x}-\mathbf{x}_n)\Big)\right\rangle\right] \\
=\;& \frac{\partial}{\partial t}\left\langle \sum_{n(\text{molecules})}\mathbf{p}_n\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle - \frac{\partial}{\partial t}\sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left[\frac{1}{6}\left\langle \sum_{n(\text{molecules})}(Q'_n)_{\alpha\beta}\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle\right] \\
=\;& \sum_\alpha \hat{e}_\alpha\frac{\partial}{\partial t}\left[P_\alpha-\sum_\beta\frac{\partial Q'_{\alpha\beta}}{\partial x_\beta}\right].
\end{aligned}$$

By Eq. (6.92), we can write the above expression as

$$\sum_\alpha \hat{e}_\alpha\frac{\partial}{\partial t}\left[P_\alpha-\sum_\beta\frac{\partial Q'_{\alpha\beta}}{\partial x_\beta}\right] = \sum_\alpha \hat{e}_\alpha\frac{\partial}{\partial t}\left[D_\alpha-\epsilon_0E_\alpha\right] = \frac{\partial}{\partial t}\left[\mathbf{D}-\epsilon_0\mathbf{E}\right].$$

Now, the average current density becomes

$$\begin{aligned}
\langle \mathbf{j}(\mathbf{x},t)\rangle &= \mathbf{j}_{\text{free}}(\mathbf{x},t)+\mathbf{j}_{\text{bound}}(\mathbf{x},t) \\
&= \left\langle \sum_{j(\text{free})} q_j\mathbf{v}_j\delta(\mathbf{x}-\mathbf{x}_j)\right\rangle + \left\langle \sum_{n(\text{molecules})} q_n\mathbf{v}_n\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle \\
&+ \frac{\partial}{\partial t}\left[\mathbf{D}(\mathbf{x},t)-\epsilon_0\mathbf{E}(\mathbf{x},t)\right]+\nabla\times\mathbf{M}(\mathbf{x},t) \\
&+ \sum_{\alpha\beta}\hat{e}_\alpha\frac{\partial}{\partial x_\beta}\left\langle \sum_{n(\text{molecules})}\Big((\mathbf{p}_n)_\alpha(\mathbf{v}_n)_\beta-(\mathbf{v}_n)_\alpha(\mathbf{p}_n)_\beta\Big)\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle \\
&- \frac{1}{6}\sum_{\alpha\beta\gamma}\hat{e}_\alpha\frac{\partial^2}{\partial x_\beta\partial x_\gamma}\left\langle \sum_{n(\text{molecules})}\Big((Q'_n)_{\alpha\beta}(\mathbf{v}_n)_\gamma-(Q'_n)_{\gamma\beta}(\mathbf{v}_n)_\alpha\Big)\delta(\mathbf{x}-\mathbf{x}_n)\right\rangle,
\end{aligned}$$

which is Eq. (6.96), with the macroscopic magnetization defined as

$$\mathbf{M}(\mathbf{x},t) = \left\langle \sum_{n(\text{molecules})} \mathbf{m}_n \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle.$$

Also, the first line gives the macroscopic current density

$$\mathbf{J}(\mathbf{x},t) = \left\langle \sum_{j(\text{free})} q_j \mathbf{v}_j \delta(\mathbf{x}-\mathbf{x}_j) + \sum_{n(\text{molecules})} q_n \mathbf{v}_n \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle.$$

(b) The last two terms in Eq. (6.96) can be combined to give

$$\begin{aligned}
& \sum_\beta \frac{\partial}{\partial x_\beta} \left\langle \sum_{n(\text{molecules})} \Big((\mathbf{p}_n)_\alpha (\mathbf{v}_n)_\beta - (\mathbf{v}_n)_\alpha (\mathbf{p}_n)_\beta \Big) \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
& -\frac{1}{6} \sum_{\beta\gamma} \frac{\partial^2}{\partial x_\beta \partial x_\gamma} \left\langle \sum_{n(\text{molecules})} \Big((Q'_n)_{\alpha\beta} (\mathbf{v}_n)_\gamma - (Q'_n)_{\beta\gamma} (\mathbf{v}_n)_\alpha \Big) \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
= \quad & \sum_\beta \frac{\partial}{\partial x_\beta} \left\langle \sum_{n(\text{molecules})} \Big((\mathbf{p}_n)_\alpha (\mathbf{v}_n)_\beta - (\mathbf{v}_n)_\alpha (\mathbf{p}_n)_\beta \Big) \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
& -\sum_\beta \frac{\partial}{\partial x_\beta} \left\langle \sum_{n(\text{molecules})} \left(\frac{1}{6} \sum_\gamma \frac{\partial (Q'_n)_{\alpha\gamma}}{\partial x_\gamma} (\mathbf{v}_n)_\beta - \frac{1}{6} \sum_\gamma \frac{\partial (Q'_n)_{\beta\gamma}}{\partial x_\gamma} (\mathbf{v}_n)_\alpha \right) \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
= \quad & \sum_\beta \frac{\partial}{\partial x_\beta} \left\langle \sum_{n(\text{molecules})} \left[(\mathbf{p}_n)_\alpha - \frac{1}{6} \sum_\gamma \frac{\partial (Q'_n)_{\alpha\gamma}}{\partial x_\gamma} \right] (\mathbf{v}_n)_\beta \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
& -\sum_\beta \frac{\partial}{\partial x_\beta} \left\langle \sum_{n(\text{molecules})} \left[(\mathbf{p}_n)_\beta - \frac{1}{6} \sum_\gamma \frac{\partial (Q'_n)_{\beta\gamma}}{\partial x_\gamma} \right] (\mathbf{v}_n)_\alpha \delta(\mathbf{x}-\mathbf{x}_n) \right\rangle \\
= \quad & \sum_\beta \frac{\partial}{\partial x_\beta} \left[P_\alpha(\mathbf{x},t) - \sum_\gamma \frac{\partial Q'_{\alpha\gamma}(\mathbf{x},t)}{\partial x_\gamma} \right] (\mathbf{v}_n)_\beta - \sum_\beta \frac{\partial}{\partial x_\beta} \left[P_\beta(\mathbf{x},t) - \sum_\gamma \frac{\partial Q'_{\beta\gamma}(\mathbf{x},t)}{\partial x_\gamma} \right] (\mathbf{v}_n)_\alpha .
\end{aligned}$$

From Eq. (6.92), we know

$$D_\alpha - \epsilon_0 E_\alpha = P_\alpha - \sum_\beta \frac{\partial Q'_{\alpha\beta}(\mathbf{x},t)}{\partial x_\beta},$$

then we can write the above expression as

$$\begin{aligned}
& \sum_\beta \frac{\partial}{\partial x_\beta} \Big[(\mathbf{D}-\epsilon_0\mathbf{E})_\alpha (\mathbf{v}_n)_\beta - (\mathbf{D}-\epsilon_0\mathbf{E})_\beta (\mathbf{v}_n)_\alpha \Big] \\
= \quad & \sum_\beta \frac{\partial}{\partial x_\beta} \epsilon_{\alpha\beta\gamma} \Big[(\mathbf{D}-\epsilon_0\mathbf{E}) \times \mathbf{v}_n \Big]_\gamma = \Big\{ \nabla \times \Big[(\mathbf{D}-\epsilon_0\mathbf{E}) \times \mathbf{v}_n \Big] \Big\}_\alpha .
\end{aligned}$$

Use this in the expression for $\langle \mathbf{j}(\mathbf{x},t) \rangle$ and plug into the second equation of Eq. (6.70), we will get

$$\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J} + \frac{\partial}{\partial t} \Big[(\mathbf{D}-\epsilon_0\mathbf{E}) \Big] + \nabla \times \mathbf{M} + \nabla \times \Big[(\mathbf{D}-\epsilon_0\mathbf{E}) \times \mathbf{v}_n \Big].$$

Rearrange terms, we will have

$$\nabla \times \left[\frac{1}{\mu_0}\mathbf{B} - \mathbf{M} - (\mathbf{D} - \epsilon_0 \mathbf{E}) \times \mathbf{v}_n \right] - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}.$$

We can define the magnetic field $\mathbf{H}$ as

$$\mathbf{H} = \frac{1}{\mu_0}\mathbf{B} - \mathbf{M} - (\mathbf{D} - \epsilon_0 \mathbf{E}) \times \mathbf{v}_n.$$

For uniform bulk motion, *i.e.*, $\mathbf{v}_n \equiv \mathbf{v}$, we finally get

$$\frac{1}{\mu_0}\mathbf{B} - \mathbf{H} = \mathbf{M} + (\mathbf{D} - \epsilon_0 \mathbf{E}) \times \mathbf{v}.$$